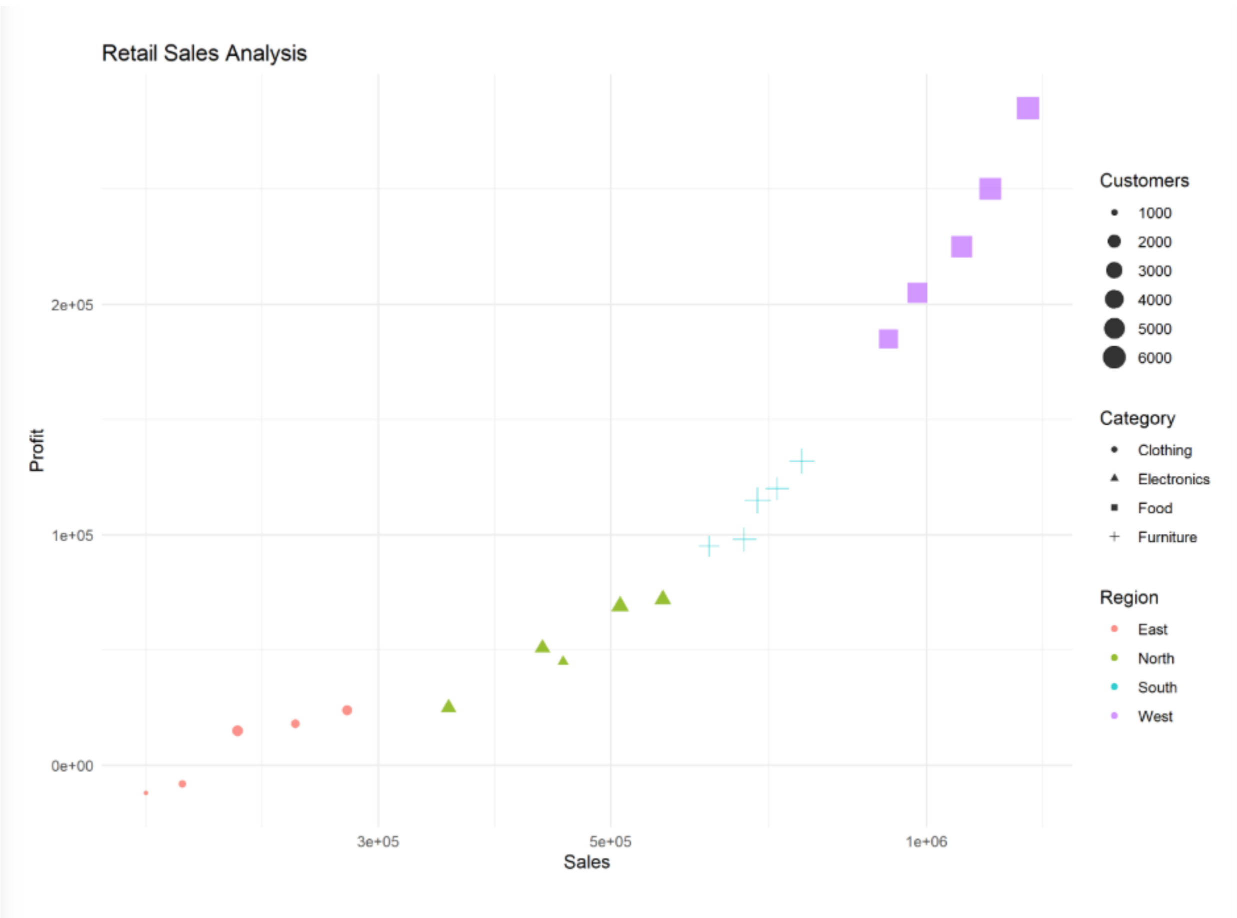
Question-1

```
library(ggplot2)

retail <- data.frame(
Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),
Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

ggplot(retail,
    aes(x = Sales,
      y = Profit,
```

```r
        size = Customers,
        color = Region,
        shape = Category)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  labs(
    title = "Retail Sales Analysis",
    x = "Sales",
    y = "Profit"
  ) +
  theme_minimal()
```



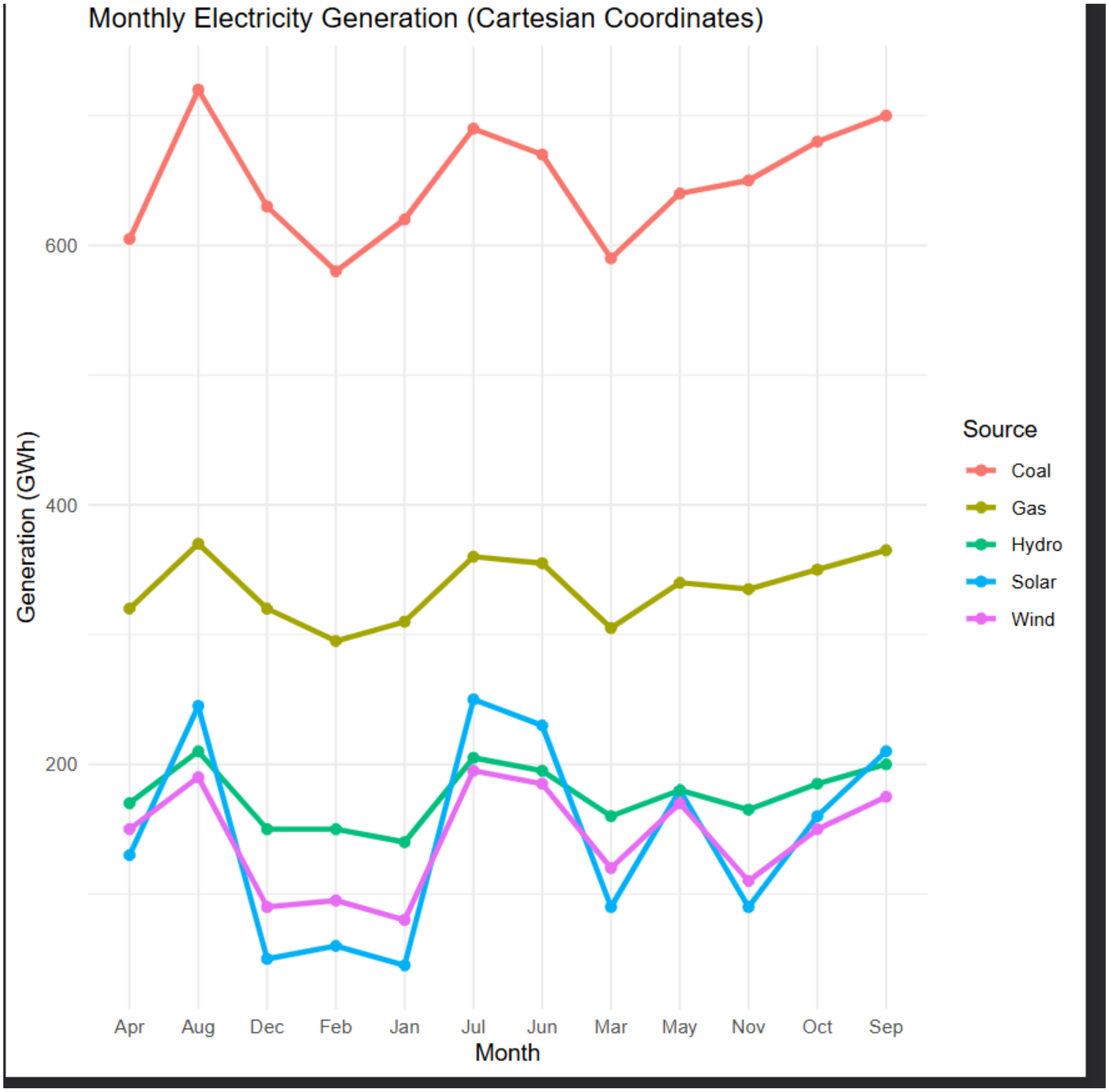
Why Bubble Chart?

- Simultaneously displays **5 variables** in one graph.
- Reveals the relationship between **Sales** and **Profit**.
- Bubble size represents **Customers**.
- Color distinguishes **Product Category**.
- Shape differentiates **Region**.
- Log scale improves visualization because Sales values span a large range.

Question-2

```
library(ggplot2)
library(tidyr)

energy <- data.frame(
  Month=month.abb,
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)

energy_long <- pivot_longer(
  energy,
  cols = -Month,
  names_to = "Source",
  values_to = "Generation"
)

ggplot(energy_long,
```

```
      aes(x = Month, y = Generation,
          color = Source, group = Source)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Monthly Electricity Generation (Cartesian Coordinates)",
    x = "Month",
    y = "Generation (GWh)"
  ) +
  theme_minimal()


ggplot(energy_long,
      aes(x = Month, y = Generation,
          fill = Source)) +
  geom_bar(stat = "identity", position = "dodge") +
  coord_polar() +
  labs(
    title = "Monthly Electricity Generation (Polar Coordinates)"
  ) +
  theme_minimal()
```



Which Coordinate System is Better?

Cartesian Coordinates provide better analytical insight because:

- Trends across months are easier to observe.
- Comparison between energy sources is more accurate.
- Exact values are easier to interpret.
- Widely used for time-series analysis.

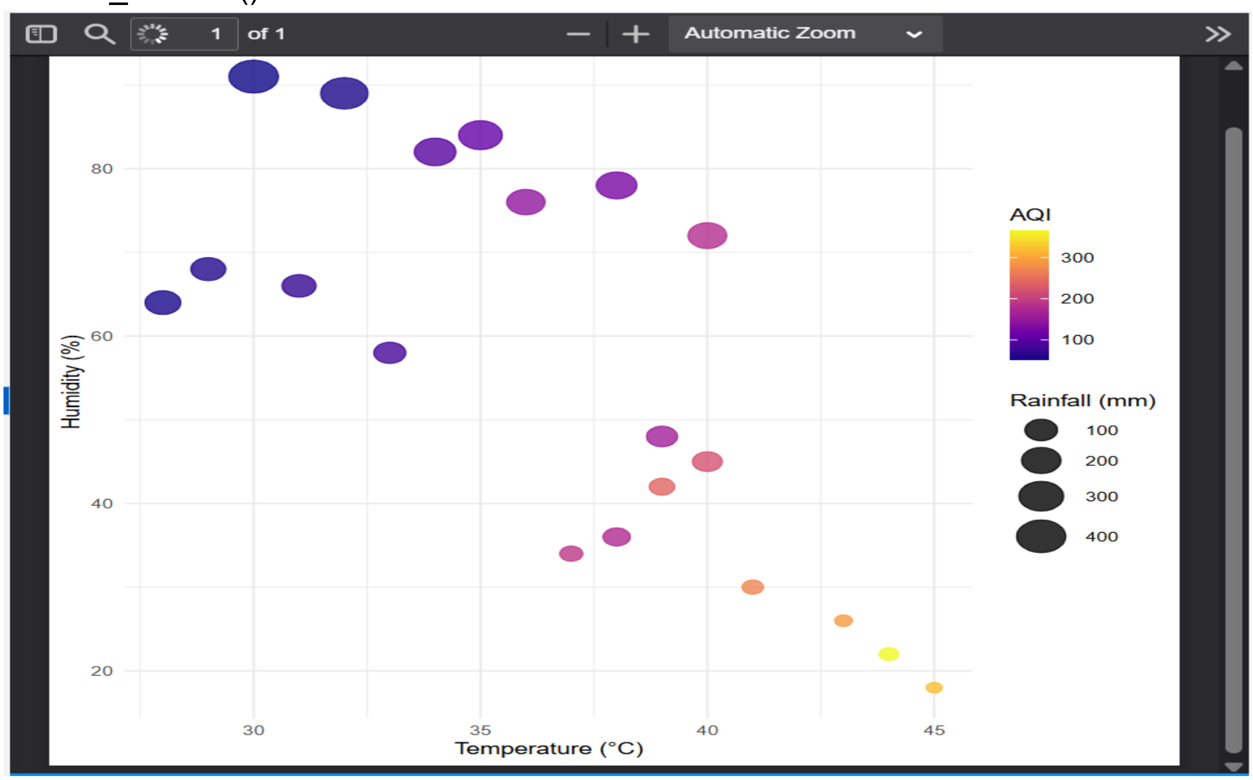
Polar Coordinates are useful mainly for:

- Displaying seasonal or cyclic patterns.
- Presentation and visual appeal rather than precise analysis.

Question-3

```
library(ggplot2)

climate <- data.frame(
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
         "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
         "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
         "Surat","Visakhapatnam","Coimbatore","Mysuru"),

  Temperature=c(44,35,38,40,39,31,33,43,45,39,
                38,40,32,30,41,37,36,34,29,28),
  Humidity=c(22,84,78,72,48,66,58,26,18,42,
             36,45,89,91,30,34,76,82,68,64),
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,
             48,72,360,410,12,20,180,240,130,145),
  AQI=c(365,110,125,180,160,72,84,290,320,240,
        175,220,58,52,270,190,145,98,60,55)
)

ggplot(climate,
       aes(x = Temperature,
           y = Humidity,
           size = Rainfall,
```

```
        colour = AQI)) +
  geom_point(alpha = 0.8) +
  scale_colour_viridis_c(option = "plasma") +
  scale_size(range = c(3,10)) +
  labs(
    title = "Climate Analysis of Major Indian Cities",
    x = "Temperature (°C)",
    y = "Humidity (%)",
    colour = "AQI",
    size = "Rainfall (mm)"
  ) +
  theme_minimal()
```



Justification of Colour Palette

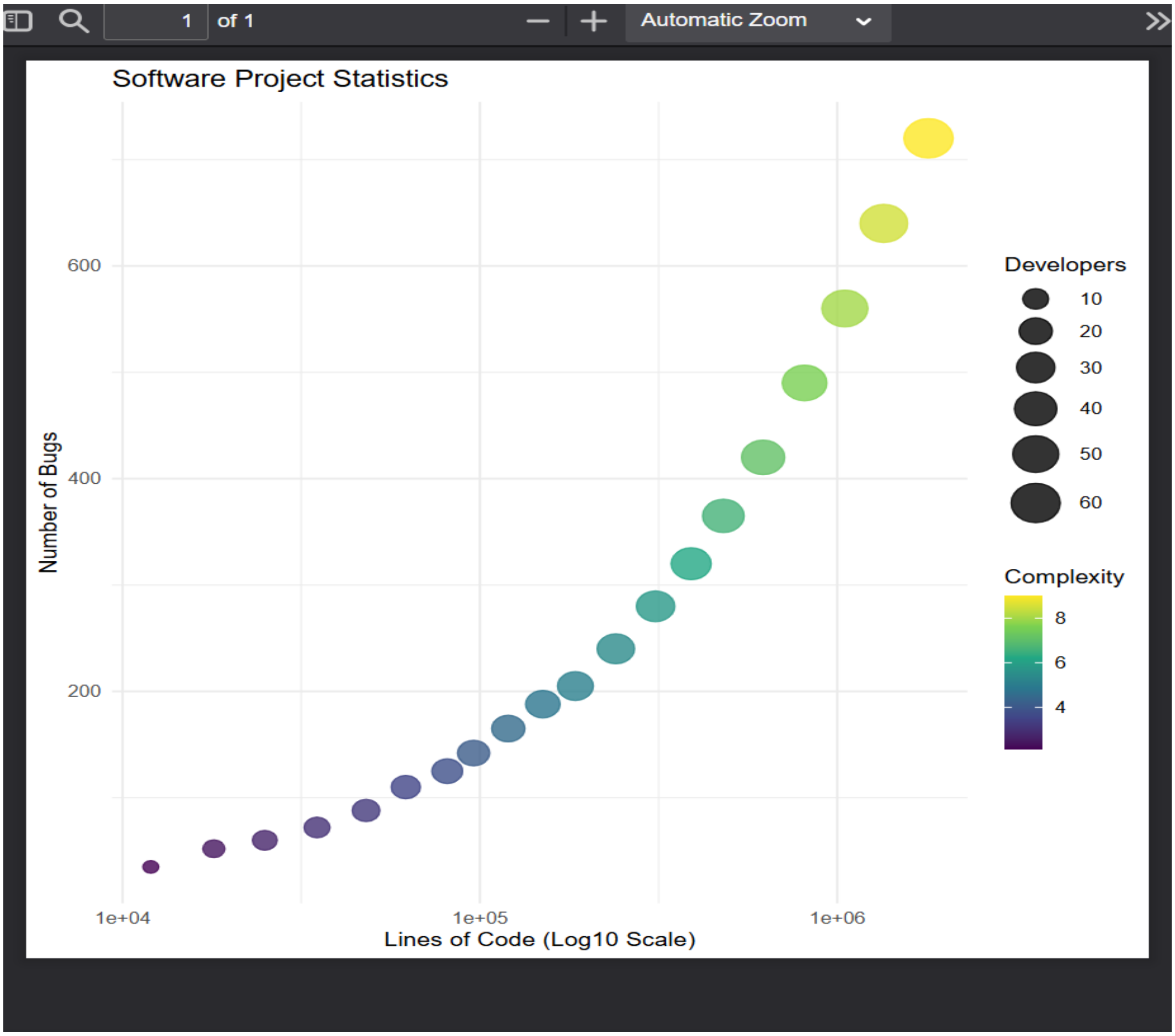
- **AQI is a continuous variable**, so a continuous colour gradient is appropriate.

- **Viridis/Plasma palette** provides smooth colour transitions from low to high AQI.
- It is **colour-blind friendly**, making the visualization accessible.
- Maintains good contrast in both colour and grayscale printing.
- Prevents misleading interpretation caused by poor colour choices.
- Widely recommended for scientific and environmental visualizations

Question-4

```
library(ggplot2)


software <- data.frame(
  Team=paste("T",1:20,sep=""),
  LinesOfCode=c(12000,18000,25000,35000,48000,
                62000,81000,96000,120000,150000,
                185000,240000,310000,390000,480000,
                620000,810000,1050000,1350000,1800000),
  Bugs=c(35,52,60,72,88,
         110,125,142,165,188,
         205,240,280,320,365,
         420,490,560,640,720),
  Developers=c(5,7,9,10,12,
               14,16,18,20,22,
               25,28,30,34,38,
               42,46,50,55,60),
  Complexity=c(2.1,2.4,2.7,3.0,3.2,
               3.5,3.8,4.1,4.5,4.8,
               5.1,5.4,5.8,6.2,6.6,
               7.1,7.6,8.0,8.5,9.0)
)
```

```
ggplot(software,
       aes(x = LinesOfCode,
           y = Bugs,
           size = Developers,
           colour = Complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  scale_colour_viridis_c() +
  scale_size(range = c(3,10)) +
  labs(
    title = "Software Project Statistics",
    x = "Lines of Code (Log10 Scale)",
    y = "Number of Bugs",
    colour = "Complexity",
    size = "Developers"
  ) +
  theme_minimal()
```



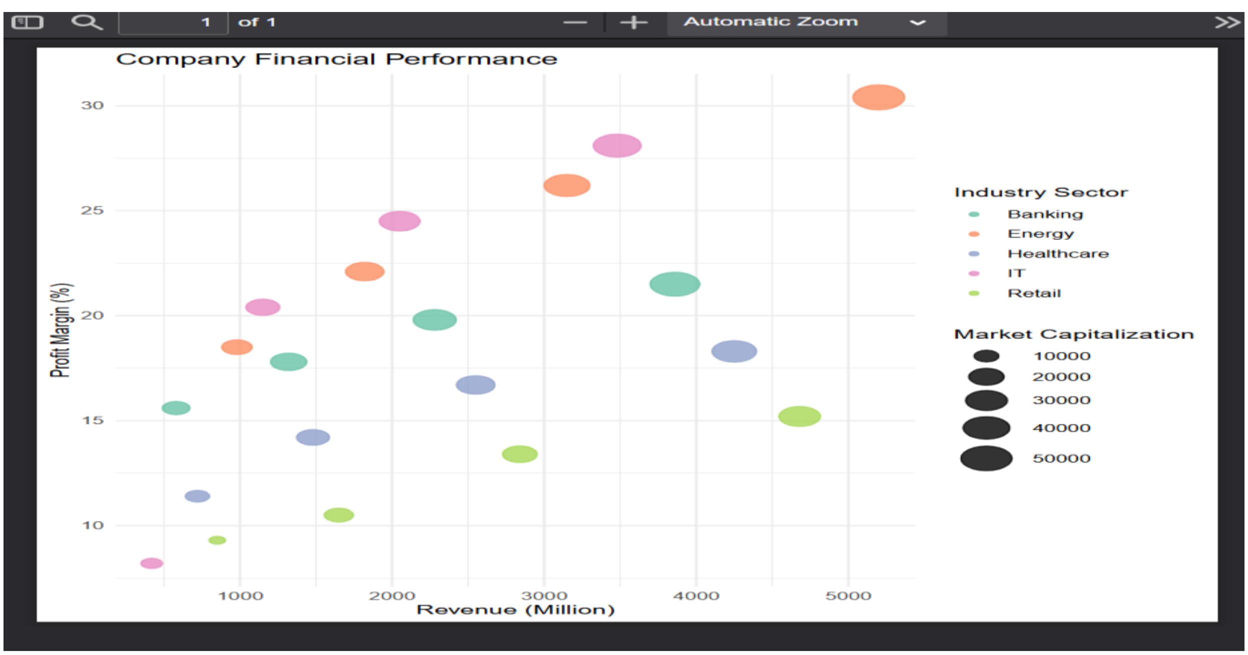
Why Log Scale is Better than Linear Scale?

Linear Scale	**Logarithmic Scale**
Small values overlap	Small values are clearly separated
Large values dominate the graph	Large values are compressed
Hard to identify trends	Trends become easier to observe
Poor readability for skewed data	Better readability and comparison

Question-5

```
library(ggplot2)


companies <- data.frame(
  Company=paste("C",1:20,sep=""),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,
            1150,1320,1480,1650,1820,
            2050,2280,2550,2840,3150,
            3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
                 20.4,17.8,14.2,10.5,22.1,
                 24.5,19.8,16.7,13.4,26.2,
                 28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800,
              5100,6800,3900,7200,4500,
              8100,9500,6200,10300,7200,
              11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,
              18200,21500,16800,13200,24500,
              28500,32500,24800,19200,38200,
              42500,46800,35600,29800,52000)
)
```

```
ggplot(companies,
       aes(x = Revenue,
           y = ProfitMargin,
           size = MarketCap,
           colour = Sector)) +
  geom_point(alpha = 0.8) +
  scale_size(range = c(3,10)) +
  scale_colour_brewer(palette = "Set2") +
  labs(
    title = "Company Financial Performance",
    x = "Revenue (Million)",
    y = "Profit Margin (%)",
    size = "Market Capitalization",
    colour = "Industry Sector"
  ) +
  theme_minimal()
```



Scale Selection

Scale	Purpose
Linear Scale	Revenue and Profit Margin have moderate ranges, so linear scale is sufficient.
Bubble Size Scale	Represents Market Capitalization.
Colour Scale	Differentiates Industry Sectors clearly.

Why This Visualization?

- Simultaneously represents Revenue, Profit Margin, Market Capitalization, and Industry Sector.
- Makes it easy to identify high-revenue and high-profit companies.
- Bubble size highlights company value (Market Capitalization).
- Different colours allow quick comparison across industry sectors.
- The clean layout minimizes perceptual bias and improves interpretation.